claire_model

Load data

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(mgcv)
```

```
Loading required package: nlme
```

```
Attaching package: 'nlme'
```

```
The following object is masked from 'package:dplyr':

    collapse
```

```
This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
train <- read_csv("data-train.csv")
```

```
Rows: 89 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): St, Re, Fr, R_moment_1, R_moment_2, R_moment_3, R_moment_4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
train <- train |>
  mutate(
    m1 = R_moment_1, m2 = R_moment_2,
    m3 = R_moment_3, m4 = R_moment_4,

    # Central moments
    C_moment_2 = m2 - m1^2,
    C_moment_3 = m3 - 3*m1*m2 + 2*m1^3,
    C_moment_4 = m4 - 4*m1*m3 + 6*m1^2*m2 - 3*m1^4,

    # Summary statistics
    sd_R = sqrt(C_moment_2),
    skewness = C_moment_3 / sd_R^3,
    kurtosis = C_moment_4 / sd_R^4,
    ex_kurt = kurtosis - 3,

    # Keep infinite Fr as a valid category
    Fr = as.factor(Fr),
    Re = as.factor(Re)
  )
```

Mean spline model

```
model_mean <- gam(
  log(R_moment_1) ~ Re + Fr + s(St, k = 5),
  data = train,
  method = "REML"
)
```

```r
summary(model_mean)
```

```
Family: gaussian
Link function: identity

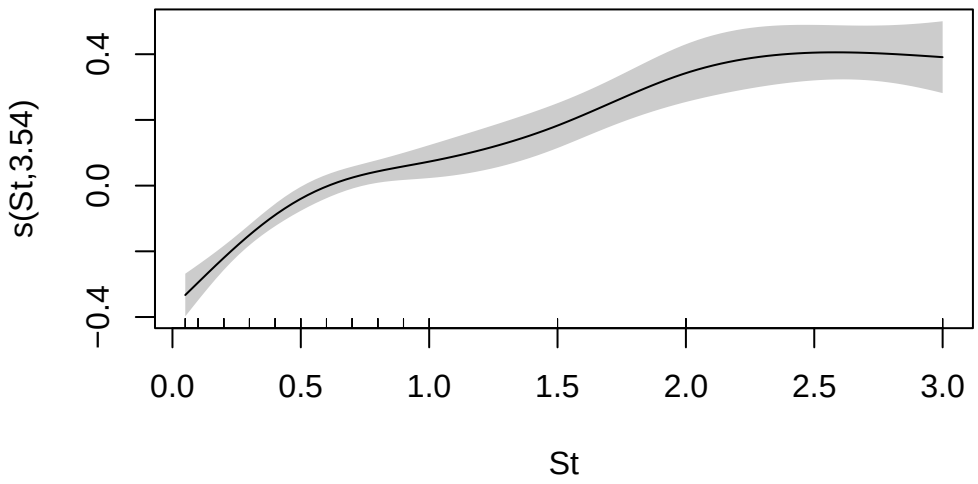
Formula:
log(R_moment_1) ~ Re + Fr + s(St, k = 5)

Parametric coefficients:
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) -2.18389    0.03144  -69.466  < 2e-16 ***
Re224       -3.63949    0.03363 -108.213  < 2e-16 ***
Re398       -5.77972    0.04254 -135.864  < 2e-16 ***
Fr0.3       -0.11795    0.03871   -3.047  0.00313 **
FrInf       -0.02276    0.03450   -0.660  0.51138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

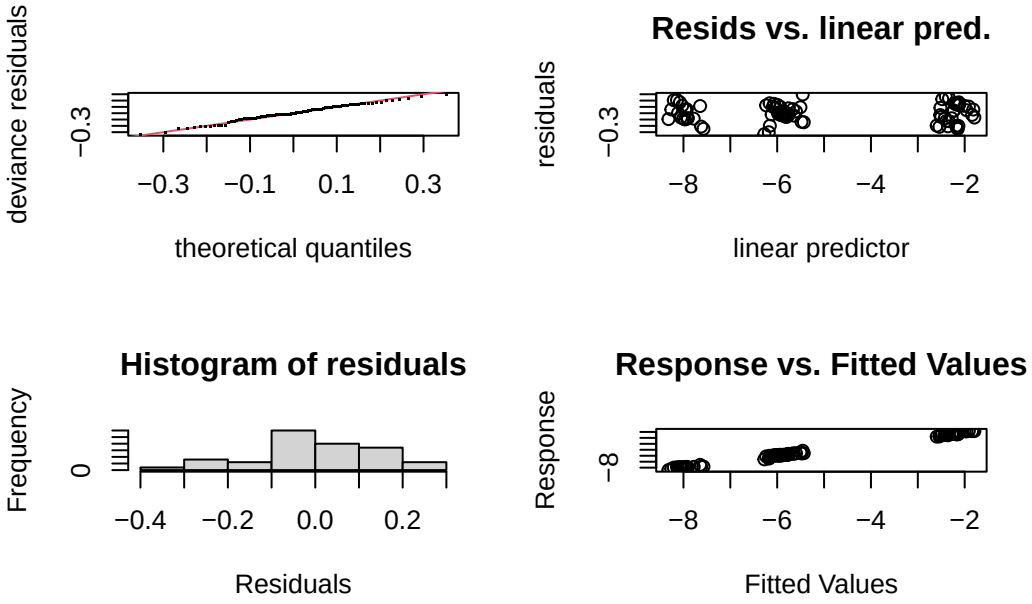
Approximate significance of smooth terms:
        edf Ref.df     F p-value
s(St) 3.537  3.859 50.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.996   Deviance explained = 99.6%
-REML = -31.975  Scale est. = 0.019356  n = 89
```

```r
plot(model_mean, shade = TRUE)
```




```
gam.check(model_mean)
```



```
Method: REML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-6.099059e-06,2.025272e-07]
(score -31.97488 & scale 0.01935551).
Hessian positive definite, eigenvalue range [0.5642027,41.53933].
Model rank =  9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

        k'  edf k-index p-value
s(St) 4.00 3.54    0.79   0.025 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Standard deviation spline model

```r
model_sd <- gam(
  log(sd_R) ~ Re + Fr + s(St, k = 5),
  data = train,
  method = "REML"
)
```

```r
summary(model_sd)
```

```
Family: gaussian
Link function: identity

Formula:
log(sd_R) ~ Re + Fr + s(St, k = 5)

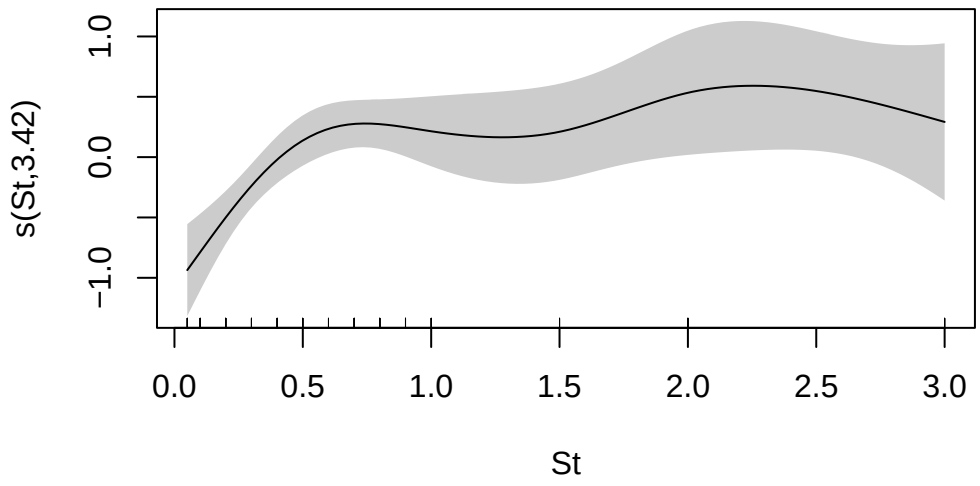
Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.9579     0.1879  10.420  < 2e-16 ***
Re224        -2.3652     0.2010 -11.767  < 2e-16 ***
Re398        -3.9885     0.2541 -15.699  < 2e-16 ***
Fr0.3        -1.7622     0.2314  -7.615  4.4e-11 ***
FrInf        -1.3763     0.2062  -6.674  2.9e-09 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

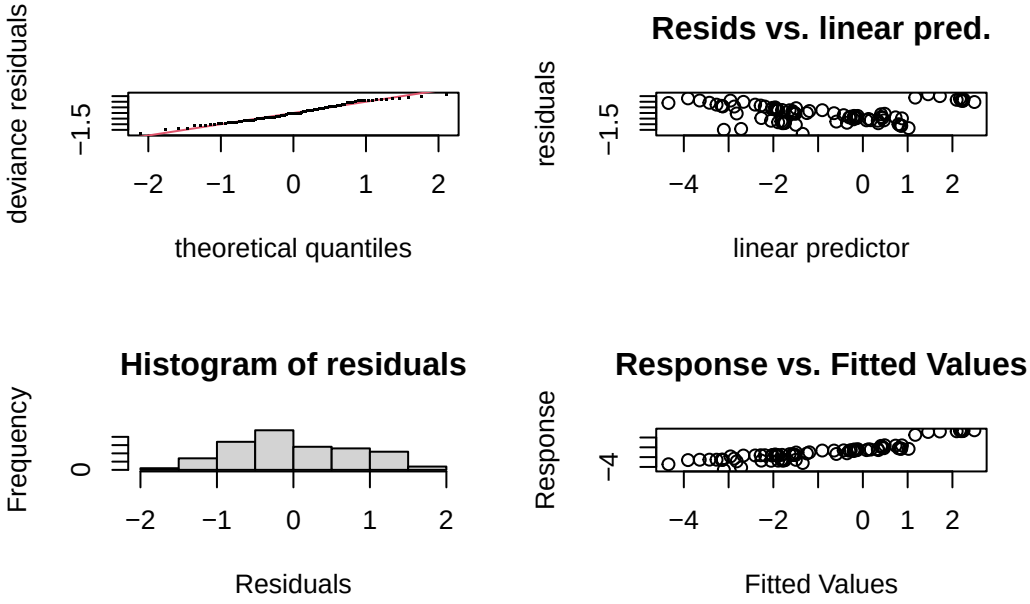
Approximate significance of smooth terms:
        edf Ref.df     F  p-value
s(St) 3.423  3.787 5.968 0.000275 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.799   Deviance explained = 81.6%
-REML = 116.01  Scale est. = 0.69155   n = 89
```

```r
plot(model_sd, shade = TRUE)
```




```r
gam.check(model_sd)
```




```
Method: REML   Optimizer: outer newton
full convergence after 4 iterations.
Gradient range [-7.049749e-06,-4.32155e-06]
(score 116.0126 & scale 0.6915505).
Hessian positive definite, eigenvalue range [0.4244224,41.53573].
Model rank =  9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

        k'  edf k-index p-value
s(St) 4.00 3.42    0.93    0.17
```

Skewness spline model

```r
model_skewness <- gam(
  log(skewness) ~ Re + Fr + s(St, k = 5),
  data = train,
  method = "REML"
)
```

```r
summary(model_skewness)
```

```
Family: gaussian
Link function: identity

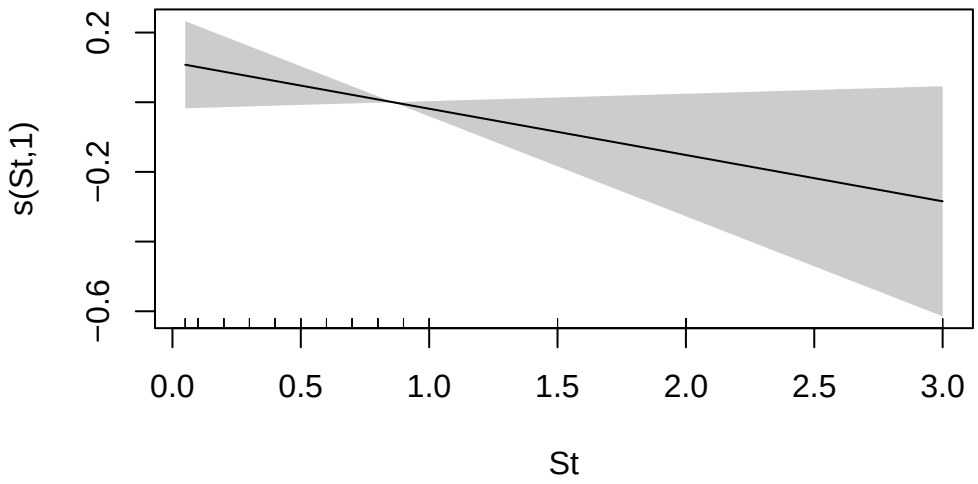
Formula:
log(skewness) ~ Re + Fr + s(St, k = 5)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.7427     0.1278  37.111  < 2e-16 ***
Re224         1.0671     0.1367   7.807 1.57e-11 ***
Re398         1.4694     0.1719   8.546 5.28e-13 ***
Fr0.3        -1.6423     0.1576 -10.423  < 2e-16 ***
FrInf        -1.3365     0.1406  -9.509 6.27e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

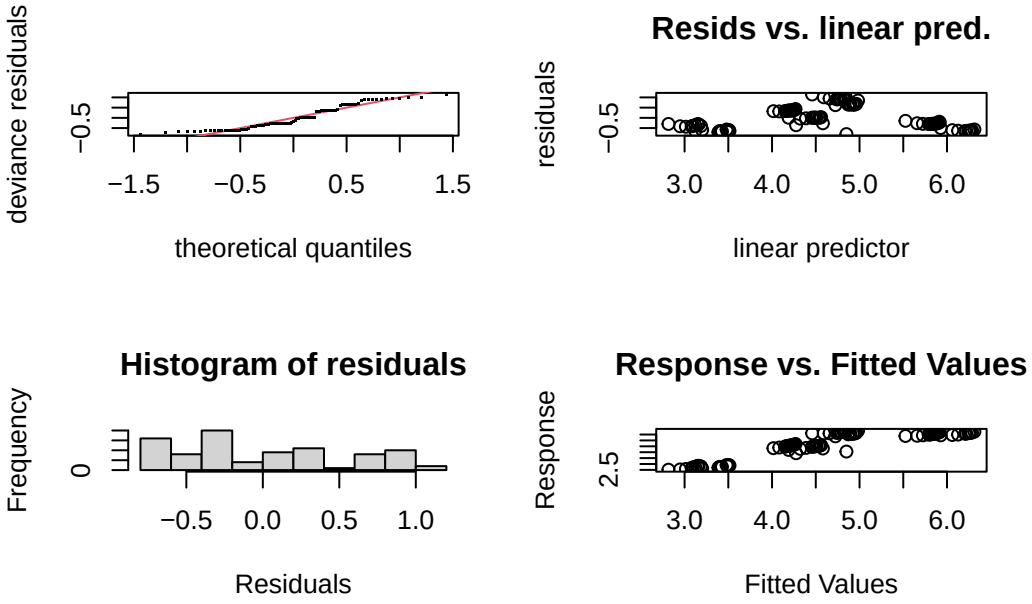
Approximate significance of smooth terms:
      edf Ref.df     F p-value
s(St)   1      1 2.967  0.0887 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.748   Deviance explained = 76.3%
-REML = 80.771  Scale est. = 0.32199   n = 89
```

```r
plot(model_skewness, shade = TRUE)
```




```
gam.check(model_skewness)
```



```
Method: REML   Optimizer: outer newton
full convergence after 8 iterations.
Gradient range [-4.525514e-05,1.140528e-05]
(score 80.77056 & scale 0.3219863).
Hessian positive definite, eigenvalue range [4.525215e-05,41.49999].
Model rank =  9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

      k' edf k-index p-value
s(St)  4   1    1.06    0.67
```

Kurtosis spline model

```r
model_kurtosis <- gam(
  log(kurtosis) ~ Re + Fr + s(St, k = 5),
  data = train,
  method = "REML"
)
```

```r
summary(model_kurtosis)
```

```
Family: gaussian
Link function: identity

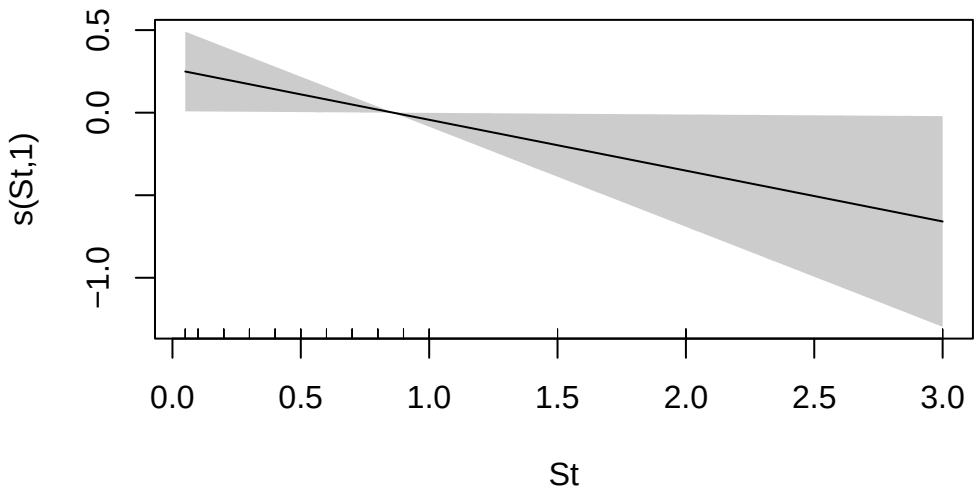
Formula:
log(kurtosis) ~ Re + Fr + s(St, k = 5)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.5622     0.2469  38.735  < 2e-16 ***
Re224         2.0730     0.2640   7.852 1.28e-11 ***
Re398         2.8498     0.3321   8.580 4.51e-13 ***
Fr0.3        -3.2577     0.3044 -10.704  < 2e-16 ***
FrInf        -2.6467     0.2715  -9.749 2.08e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

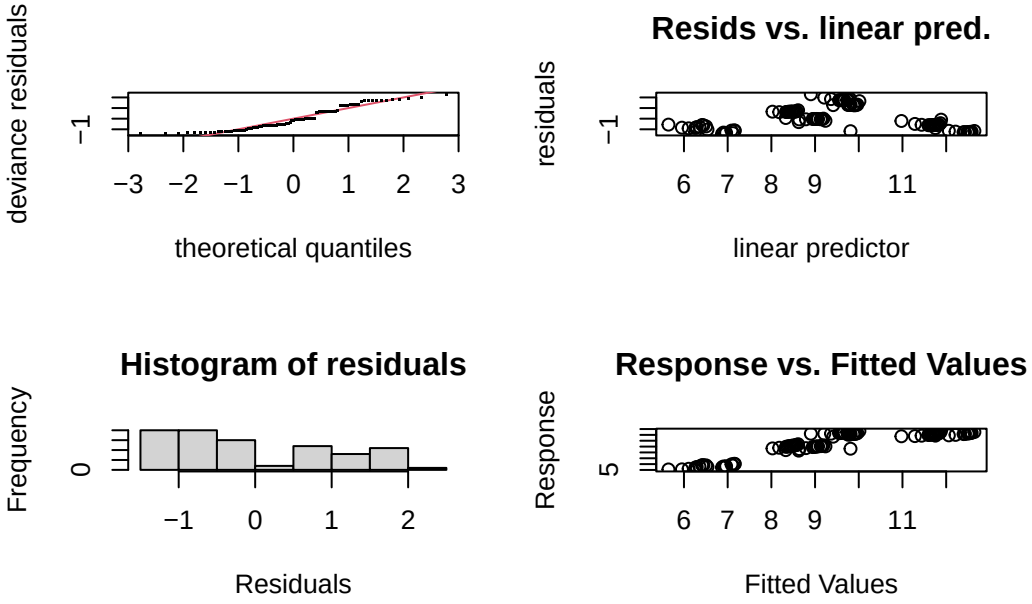
```
Approximate significance of smooth terms:
       edf Ref.df     F p-value
s(St)    1      1 4.268  0.0419 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.756   Deviance explained = 76.9%
-REML = 135.42  Scale est. = 1.2014    n = 89
```

```r
plot(model_kurtosis, shade = TRUE)
```




```r
gam.check(model_kurtosis)
```




```
Method: REML   Optimizer: outer newton
full convergence after 8 iterations.
Gradient range [-5.780349e-05,2.576094e-06]
(score 135.4156 & scale 1.201417).
Hessian positive definite, eigenvalue range [5.779822e-05,41.5].
Model rank =  9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

      k' edf k-index p-value
s(St)  4   1    1.07    0.76
```

LOOCV

```r
predict_original <- function(model, newdata) {
  smearing_factor <- mean(exp(residuals(model, type = "response")))
  exp(predict(model, newdata = newdata)) * smearing_factor
}
```

```r
loocv_predict <- function(model, data) {
  map_dbl(seq_len(nrow(data)), function(i) {
    fold_model <- update(model, data = data[-i, ])
    predict_original(fold_model, data[i, ])
  })
}

predictions <- train |>
  mutate(
    pred_mean = loocv_predict(model_mean, train),
    pred_sd = loocv_predict(model_sd, train),
    pred_skewness = loocv_predict(model_skewness, train),
    pred_kurtosis = loocv_predict(model_kurtosis, train)
  )
```

```r
score_predictions <- function(response, observed, predicted) {
  tibble(
    Response = response,
    RMSE = sqrt(mean((observed - predicted)^2)),
    MAE = mean(abs(observed - predicted))
  )
}

performance <- bind_rows(
  score_predictions("Mean", predictions$R_moment_1, predictions$pred_mean),
  score_predictions("Standard deviation", predictions$sd_R, predictions$pred_sd),
  score_predictions("Skewness", predictions$skewness, predictions$pred_skewness),
  score_predictions("Kurtosis", predictions$kurtosis, predictions$pred_kurtosis)
)
```

Prediction performance summary

```r
knitr::kable(
  performance,
  digits = c(0, 4, 5),
  format.args = list(big.mark = ","),
  caption = "Spline model performance using LOOCV"
)
```

Table 1: Spline model performance using LOOCV

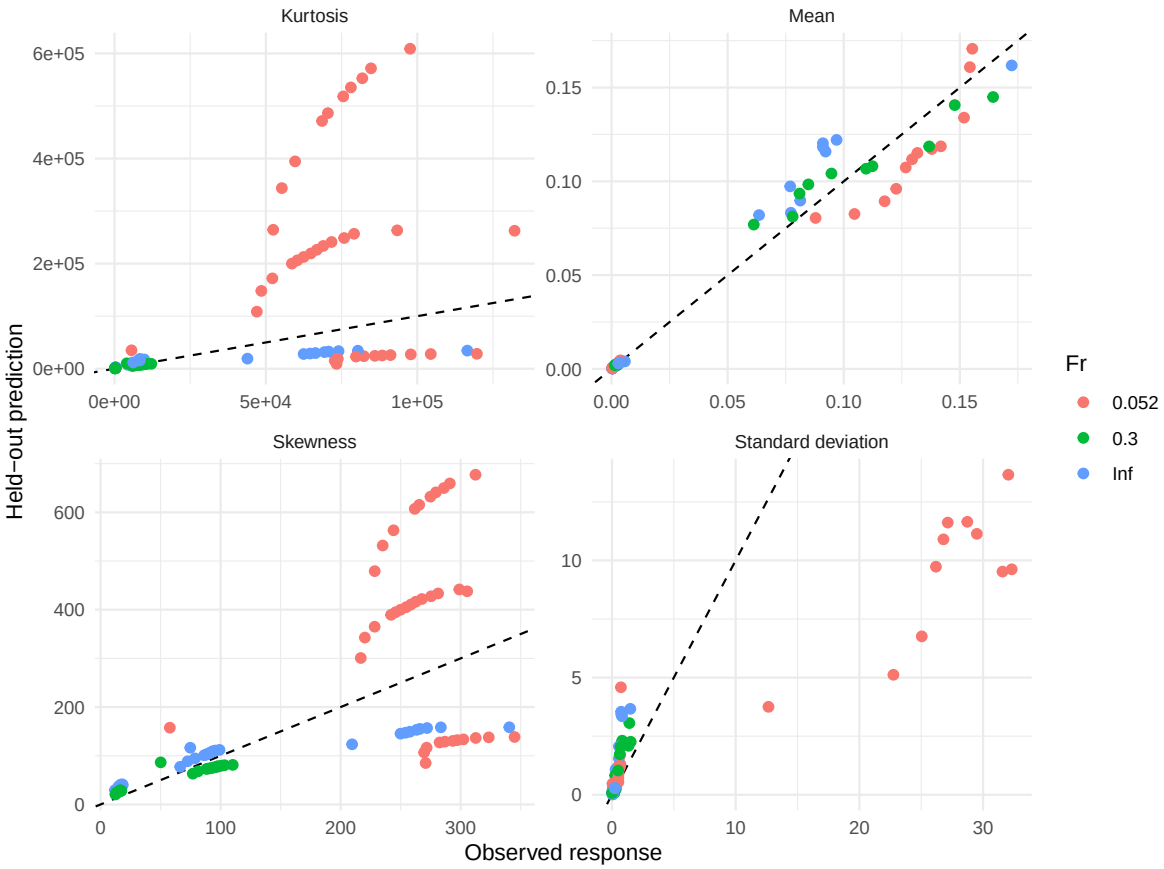
Response	RMSE	MAE
Mean	0.0105	0.00571
Standard deviation	6.2951	2.61292
Skewness	146.7178	102.92469
Kurtosis	152,604.3299	81,907.92780

Observed versus predicted: all four responses

```
compare_response <- function(label, observed, predicted) {
  tibble(
    Response = label,
    St = predictions$St, Re = predictions$Re, Fr = predictions$Fr,
    Observed = observed,
    Predicted = predicted,
    Absolute_error = abs(predicted - observed)
  )
}

comparison_data <- bind_rows(
  compare_response("Mean", predictions$R_moment_1, predictions$pred_mean),
  compare_response("Standard deviation", predictions$sd_R, predictions$pred_sd),
  compare_response("Skewness", predictions$skewness, predictions$pred_skewness),
  compare_response("Kurtosis", predictions$kurtosis, predictions$pred_kurtosis)
)
```

```
ggplot(comparison_data, aes(x = Observed, y = Predicted)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  geom_point(aes(color = Fr), size = 2) +
  facet_wrap(~Response, scales = "free") +
  labs(x = "Observed response", y = "Held-out prediction", color = "Fr") +
  theme_minimal()
```



Spline models predict mean cluster volume more successfully than the other three responses using LOOCV.

Many large skewness and kurtosis errors occur at `Fr` $= 0.052$.

Extension: separate St curves by Fr

Sd

```
model_sd_fr_train <- gam(
  log(sd_R) ~ Re + Fr + s(St, by = Fr, k = 5),
  data = train,
  method = "REML"
)

summary(model_sd_fr_train)
```

```
Family: gaussian
Link function: identity

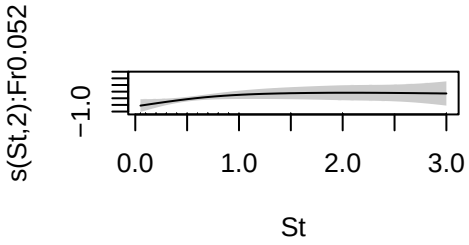
Formula:
log(sd_R) ~ Re + Fr + s(St, by = Fr, k = 5)

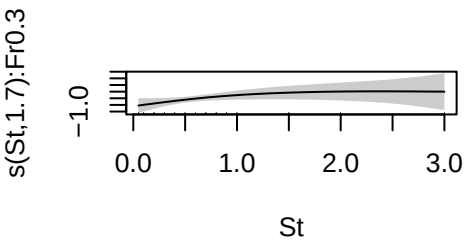
Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.9314     0.2005   9.635 5.95e-15 ***
Re224        -2.3365     0.2163 -10.804  < 2e-16 ***
Re398        -3.9177     0.2705 -14.486  < 2e-16 ***
Fr0.3        -1.7496     0.2468  -7.090 5.15e-10 ***
FrInf        -1.3846     0.2199  -6.298 1.62e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

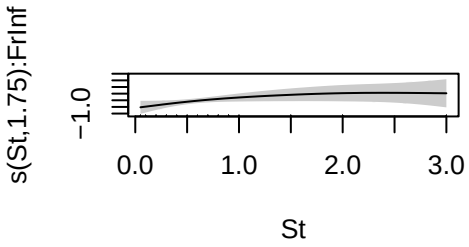
Approximate significance of smooth terms:
                edf Ref.df     F p-value
s(St):Fr0.052 1.998  2.439 2.398   0.118
s(St):Fr0.3   1.698  2.101 1.904   0.164
s(St):FrInf   1.749  2.142 2.260   0.111

R-sq.(adj) =  0.772   Deviance explained = 79.6%
-REML = 119.72  Scale est. = 0.78585   n = 89
```

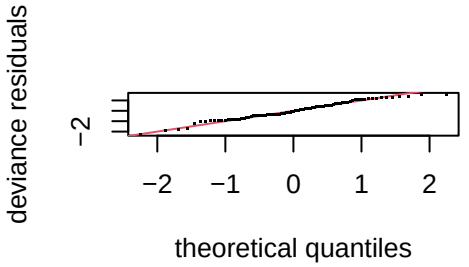
```r
plot(model_sd_fr_train, shade = TRUE, pages = 1)
```

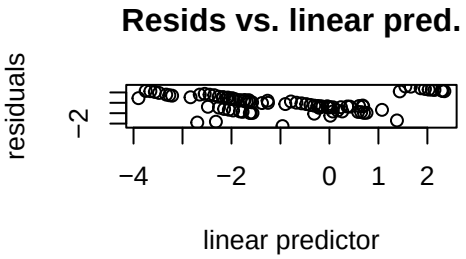


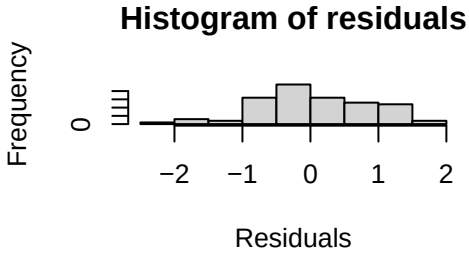


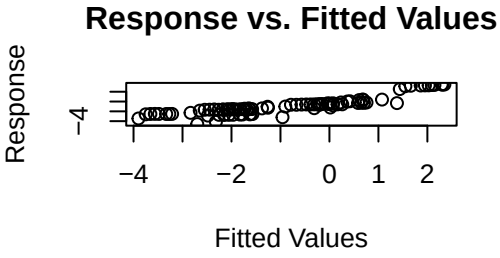



```
gam.check(model_sd_fr_train)
```









```
Method: REML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-1.04697e-07,1.837009e-09]
(score 119.7187 & scale 0.785846).
Hessian positive definite, eigenvalue range [0.1635452,40.51268].
Model rank =  17 / 17

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                k'  edf k-index p-value
s(St):Fr0.052 4.00 2.00    0.85   0.070 .
s(St):Fr0.3   4.00 1.70    0.85   0.050 *
s(St):FrInf   4.00 1.75    0.85   0.045 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
sd_fr_prediction <- loocv_predict(model_sd_fr_train, train)

sd_model_comparison <- bind_rows(
  score_predictions("Shared St curve", train$sd_R,
                    predictions$pred_sd),
  score_predictions("Fr-specific St curves", train$sd_R,
                    sd_fr_prediction)
) |>
  rename(Model = Response)

knitr::kable(sd_model_comparison, digits = 2,
             caption = "Standard deviation: baseline versus Fr-specific curves")
```

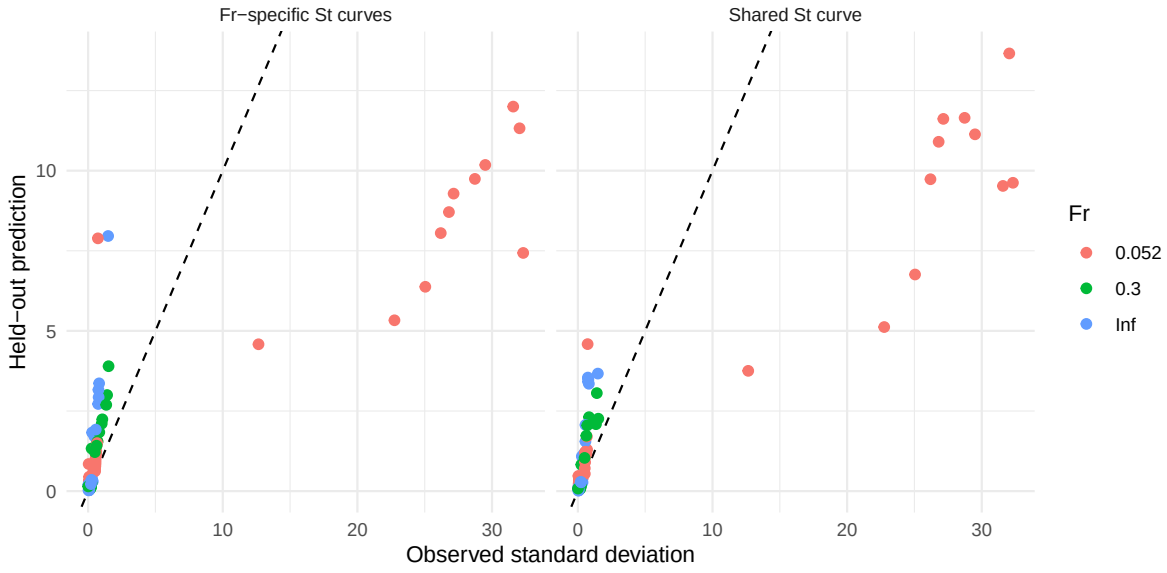
Table 2: Standard deviation: baseline versus Fr-specific curves

Model	RMSE	MAE
Shared St curve	6.3	2.61
Fr-specific St curves	6.7	2.84

```
sd_plot_data <- bind_rows(
  tibble(Model = "Shared St curve", Observed = train$sd_R,
         Predicted = predictions$pred_sd, Fr = train$Fr),
  tibble(Model = "Fr-specific St curves", Observed = train$sd_R,
         Predicted = sd_fr_prediction, Fr = train$Fr)
```

```
)

ggplot(sd_plot_data, aes(Observed, Predicted, color = Fr)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  geom_point(size = 2) +
  facet_wrap(~Model) +
  labs(x = "Observed standard deviation", y = "Held-out prediction") +
  theme_minimal()
```



Skewness

```
model_skewness_fr_train <- gam(
  log(skewness) ~ Re + Fr + s(St, by = Fr, k = 5),
  data = train,
  method = "REML"
)

summary(model_skewness_fr_train)
```

```
Family: gaussian
Link function: identity

Formula:
```

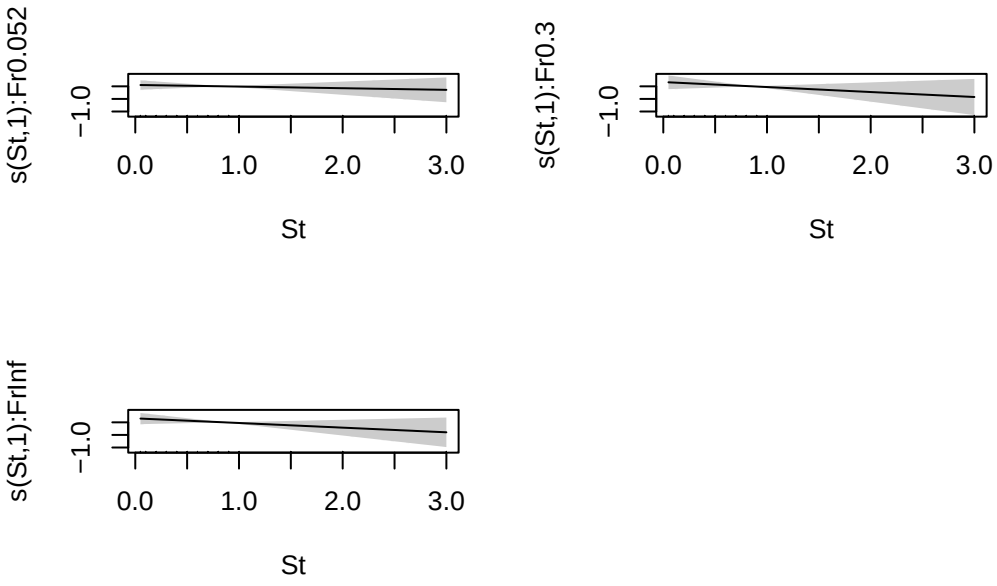
```
log(skewness) ~ Re + Fr + s(St, by = Fr, k = 5)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.7442     0.1291  36.752  < 2e-16 ***
Re224         1.0633     0.1391   7.643 3.78e-11 ***
Re398         1.4568     0.1741   8.368 1.41e-12 ***
Fr0.3        -1.6428     0.1589 -10.340  < 2e-16 ***
FrInf        -1.3351     0.1418  -9.416 1.20e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

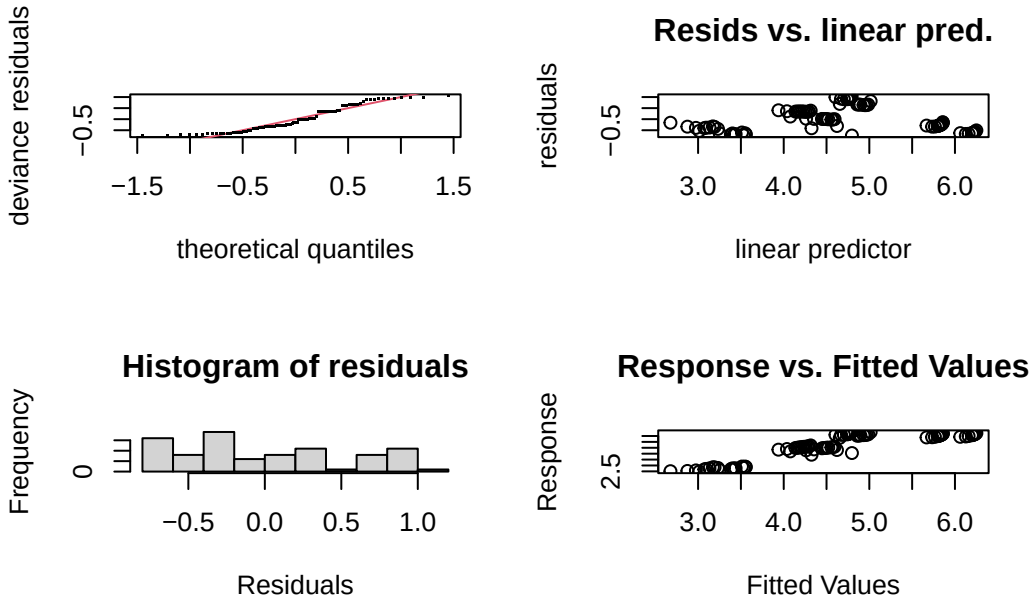
Approximate significance of smooth terms:
              edf Ref.df     F p-value
s(St):Fr0.052   1      1 0.319   0.574
s(St):Fr0.3     1      1 1.393   0.241
s(St):FrInf     1      1 1.819   0.181

R-sq.(adj) =  0.744   Deviance explained = 76.4%
-REML =  82.49  Scale est. = 0.32735   n = 89
```

```r
plot(model_skewness_fr_train, shade = TRUE, pages = 1)
```



```
gam.check(model_skewness_fr_train)
```




```
Method: REML   Optimizer: outer newton
full convergence after 10 iterations.
Gradient range [-8.964209e-06,4.001095e-06]
(score 82.49003 & scale 0.3273453).
Hessian positive definite, eigenvalue range [3.755934e-06,40.5].
Model rank =  17 / 17

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

              k' edf k-index p-value
s(St):Fr0.052  4   1    1.05    0.59
s(St):Fr0.3    4   1    1.05    0.64
s(St):FrInf    4   1    1.05    0.63
```

```
skewness_fr_prediction <- loocv_predict(model_skewness_fr_train, train)

skewness_model_comparison <- bind_rows(
```

```
  score_predictions("Shared St curve", train$skewness,
                    predictions$pred_skewness),
  score_predictions("Fr-specific St curves", train$skewness,
                    skewness_fr_prediction)
) |>
  rename(Model = Response)

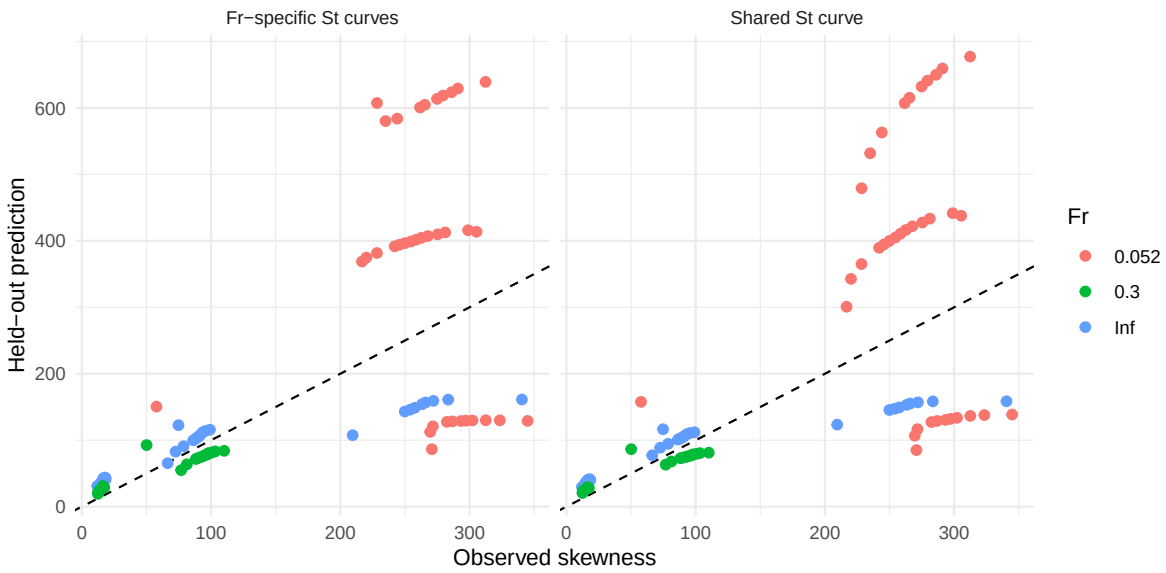
knitr::kable(skewness_model_comparison, digits = 2,
             caption = "Skewness: baseline versus Fr-specific curves")
```

Table 3: Skewness: baseline versus Fr-specific curves

Model	RMSE	MAE
Shared St curve	146.72	102.92
Fr-specific St curves	147.60	103.74

```
skewness_plot_data <- bind_rows(
  tibble(Model = "Shared St curve", Observed = train$skewness,
         Predicted = predictions$pred_skewness, Fr = train$Fr),
  tibble(Model = "Fr-specific St curves", Observed = train$skewness,
         Predicted = skewness_fr_prediction, Fr = train$Fr)
)

ggplot(skewness_plot_data, aes(Observed, Predicted, color = Fr)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  geom_point(size = 2) +
  facet_wrap(~Model) +
  labs(x = "Observed skewness", y = "Held-out prediction") +
  theme_minimal()
```



Kurtosis

```
model_kurtosis_fr_train <- gam(
  log(kurtosis) ~ Re + Fr + s(St, by = Fr, k = 5),
  data = train,
  method = "REML"
)

summary(model_kurtosis_fr_train)
```

```
Family: gaussian
Link function: identity

Formula:
log(kurtosis) ~ Re + Fr + s(St, by = Fr, k = 5)

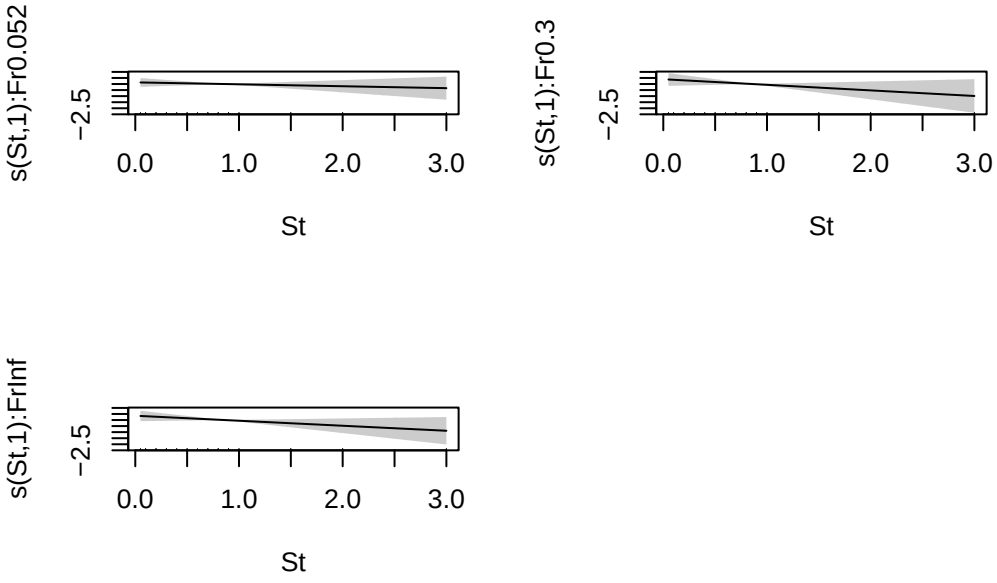
Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.5661     0.2491  38.408  < 2e-16 ***
Re224         2.0631     0.2684   7.686 3.12e-11 ***
Re398         2.8221     0.3359   8.402 1.21e-12 ***
Fr0.3        -3.2588     0.3065 -10.631  < 2e-16 ***
FrInf        -2.6434     0.2736  -9.662 3.91e-15 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

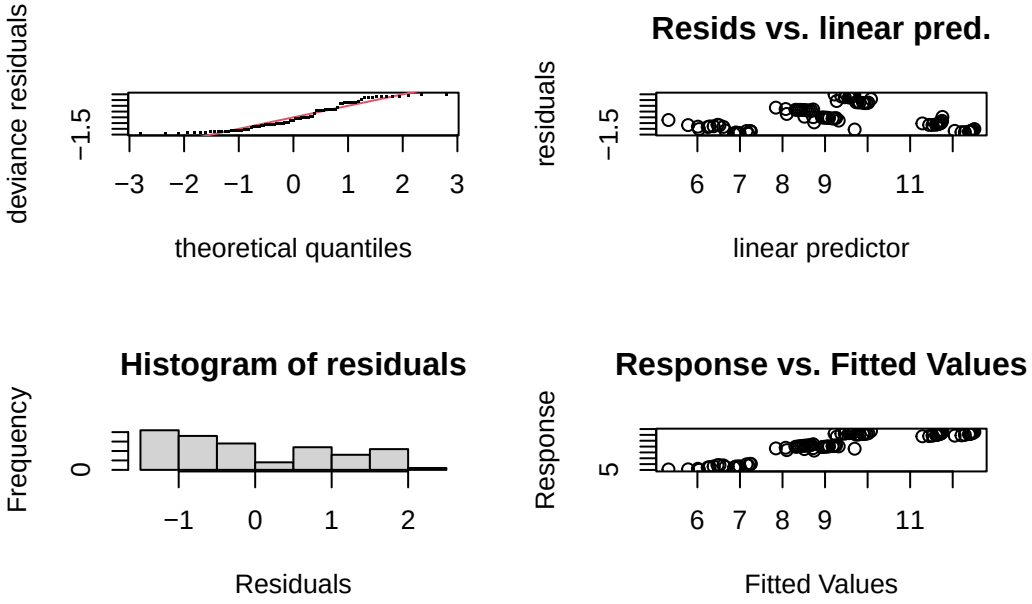
Approximate significance of smooth terms:
              edf Ref.df     F p-value
s(St):Fr0.052   1      1 0.517   0.474
s(St):Fr0.3     1      1 2.030   0.158
s(St):FrInf     1      1 2.450   0.121

R-sq.(adj) =  0.752   Deviance explained = 77.2%
-REML = 135.73  Scale est. = 1.2186    n = 89
```

```
plot(model_kurtosis_fr_train, shade = TRUE, pages = 1)
```




```
gam.check(model_kurtosis_fr_train)
```




```
Method: REML   Optimizer: outer newton
full convergence after 9 iterations.
Gradient range [-2.476023e-05,6.752467e-06]
(score 135.7253 & scale 1.218611).
Hessian positive definite, eigenvalue range [1.009076e-05,40.49999].
Model rank =  17 / 17

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

              k' edf k-index p-value
s(St):Fr0.052  4   1    1.05    0.72
s(St):Fr0.3    4   1    1.05    0.69
s(St):FrInf    4   1    1.05    0.70
```

```
kurtosis_fr_prediction <- loocv_predict(model_kurtosis_fr_train, train)

kurtosis_model_comparison <- bind_rows(
  score_predictions("Shared St curve", train$kurtosis,
                    predictions$pred_kurtosis),
  score_predictions("Fr-specific St curves", train$kurtosis,
```

```
                    kurtosis_fr_prediction)
) |>
  rename(Model = Response)

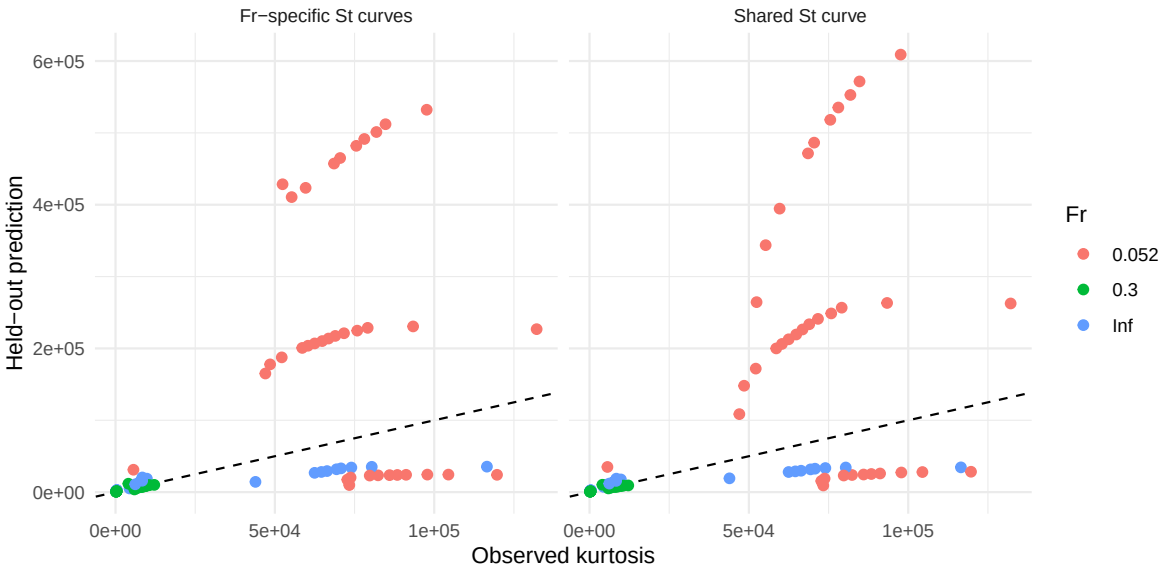
knitr::kable(kurtosis_model_comparison, digits = 2,
             caption = "Kurtosis: baseline versus Fr-specific curves")
```

Table 4: Kurtosis: baseline versus Fr-specific curves

Model	RMSE	MAE
Shared St curve	152604.3	81907.93
Fr-specific St curves	147245.7	80624.87

```
kurtosis_plot_data <- bind_rows(
  tibble(Model = "Shared St curve", Observed = train$kurtosis,
         Predicted = predictions$pred_kurtosis, Fr = train$Fr),
  tibble(Model = "Fr-specific St curves", Observed = train$kurtosis,
         Predicted = kurtosis_fr_prediction, Fr = train$Fr)
)

ggplot(kurtosis_plot_data, aes(Observed, Predicted, color = Fr)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  geom_point(size = 2) +
  facet_wrap(~Model) +
  labs(x = "Observed kurtosis", y = "Held-out prediction") +
  theme_minimal()
```



Under LOOCV, Fr-specific curves still have higher RMSE and MAE for SD and skewness. For kurtosis, they modestly improve both metrics: RMSE decreases from about 152,604 to 147,246 and MAE from 81,908 to 80,625.

- Mean predictions track observed values reasonably well.
- Large standard deviations are still underpredicted.
- Skewness and kurtosis still have substantial prediction errors.